Colter Snyder

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Hiring Team
United Launch Alliance
Centennial, CO 80112

Re: Launch Systems & Real-Time Controls Software Engineer 2, Ground Software Fluids (Req. 2255)

Dear Hiring Team,

Before a ULA rocket flies, software has to load it with cryogenic propellant safely and on time, through manufacturing, preflight test, and countdown. I would like to help write and maintain that software. I am a software engineer at Innovative Scientific Solutions, where I develop ITAR- and CUI-controlled nuclear weapons effects simulation platforms for defense organizations. I hold an M.S. in Computer Science from the Colorado School of Mines, and I would bring two years of experience building software whose results people rely on.

At ISSI I work across the full lifecycle on a small agile team and have led development of several parts of our codebase. I refine requirements, design object-oriented architectures, document them in UML and design documents, implement features in Python and Rust on Linux, and verify them with automated tests in GitLab CI/CD with SonarQube analysis. I spearheaded our internal performance-benchmarking tool, a Python command-line application with a GUI front end that drives our calculation services effect by effect, alone and in combination, and charts the results. Much of my work is porting legacy physics models from Fortran and Python to Rust without changing their numerical behavior. Both have taught me to treat every change as something that must be measured and proven correct, not just made to compile.

What draws me to this role in particular is where software meets hardware. Outside work I program in C, design FPGA logic in Verilog, and write 6502 and Z80 assembly, where timing and the physical behavior of the hardware matter as much as the code. My graduate coursework in cyber-physical systems security and IoT security focused on how software that commands physical equipment can fail or be attacked, and how to design against that. Commanding propulsion and electrical ground support equipment under hard real-time requirements is the professional version of the problems I seek out on my own time, and I am eager to deepen my work in C++ and real-time Linux on systems where it counts.

The role also calls for presenting designs and working closely with a range of stakeholders. At ISSI I take part in technical meetings with our government customers, and I co-taught CSCI 220: Data Structures and Algorithms in C++ to 120 students and taught another section of 90. Teaching taught me to explain technical designs clearly to audiences with very different backgrounds. I am comfortable with a fail-fast culture: I would rather find a problem on my bench or in a test run than on the pad.

Space has fascinated me for as long as I can remember, and contributing to ULA's record of mission success here in Colorado, close to my home in Lakewood, is an opportunity I would be proud to take. I would welcome the chance to discuss how I can support the Ground Software Fluids team. Thank you for your time and consideration.

Sincerely,

Colter Snyder